

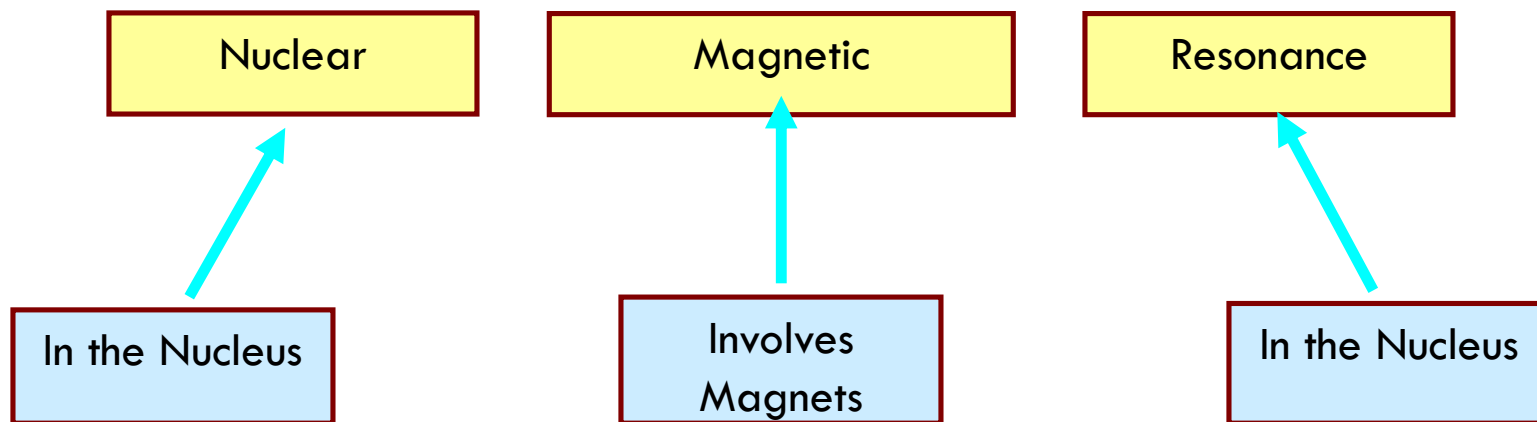
NMR SPECTROSCOPY

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Theory of NMR

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- A spectroscopic technique that gives us information about the number and types of atoms in a molecule.
- Nuclear magnetic resonance spectroscopy is a powerful analytical technique used to characterize organic molecules by identifying carbon-hydrogen frameworks within molecules.



- Two common types of NMR spectroscopy are used to characterize organic structure: ^1H NMR is used to determine the type and number of H atoms in a molecule; ^{13}C NMR is used to determine the type and number of C atoms in the molecule.
- The source of energy in NMR is radio waves which have long wavelengths, and thus low energy and frequency.
- When low-energy radio waves interact with a molecule, they can change the nuclear spins of certain atoms in presence of strong magnetic fields, including ^1H and ^{13}C .
- All the atoms contains nuclei and all nuclei contains protons (+ve) charge in which some charge nuclei posses “Spin” on their own axis.

- Spin nuclei are those which contains either **Odd** atomic number or odd atomic mass number or both e.g. ^1H , ^2H , ^{13}C , ^{14}N , ^{17}O , ^{35}Cl etc are useful for NMR.
- Those nuclei contains **Even** number of atomic and mass number are not useful for NMR e.g. ^{12}C , ^{16}O etc.
- The nuclei posses spin, they spin on their nuclear axis leads to generate magnetic dipole ' μ ' so the angular momentum of this spinning charge is quantified and described by Quantum Spin Number " I ".

Element	^1H	^2H	^{12}C	^{13}C	^{14}N	^{15}N	^{16}O	^{19}F	^{31}P	^{32}S
Nuclear spin quantum number (I)	1/2	1	0	1/2	1	1/2	0	1/2	1/2	0
Number of spin states	2	3	1	2	3	2	1	2	2	1

- The individual protons have spin quantum number $+1/2$ or $-1/2$.
- i.e. Hydrogen have spin quantum number (I) = $+1/2$ or $-1/2$.

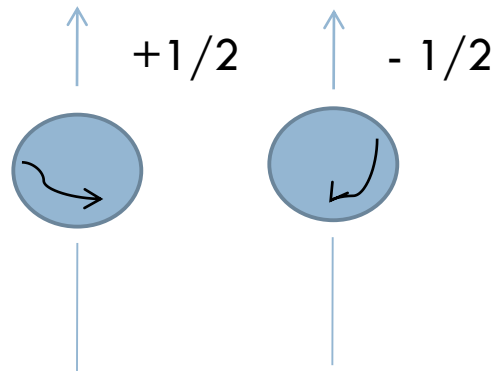


Fig. 1. Two spin states allowed for proton (H)

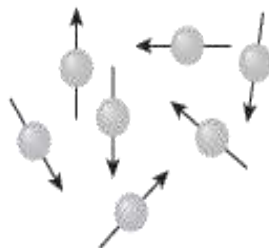
- These spin states have equal amount of energy (degenerated) in the absence of magnetic field.
- When a charged particle such as a proton spins on its axis, it creates a magnetic field. Thus, the nucleus can be considered to be a tiny bar magnet.

- But when magnetic field is applied, the proton (H) possesses spin & their own magnetic field align themselves either or opposite to magnetic field.
- For e.g. ^1H has $+1/2$ & $-1/2$ spin state, the proton (H) with $+1/2$ spin state align themselves with field (Lower energy) and with $-1/2$ spin state align opposite to field (Higher energy).

A spinning proton
creates a magnetic field.

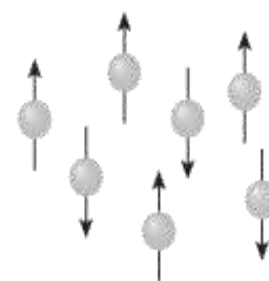


With no external magnetic field...



The nuclear magnets are
randomly oriented.

In a magnetic field...



The nuclear magnets are
oriented **with or against** B_0 .

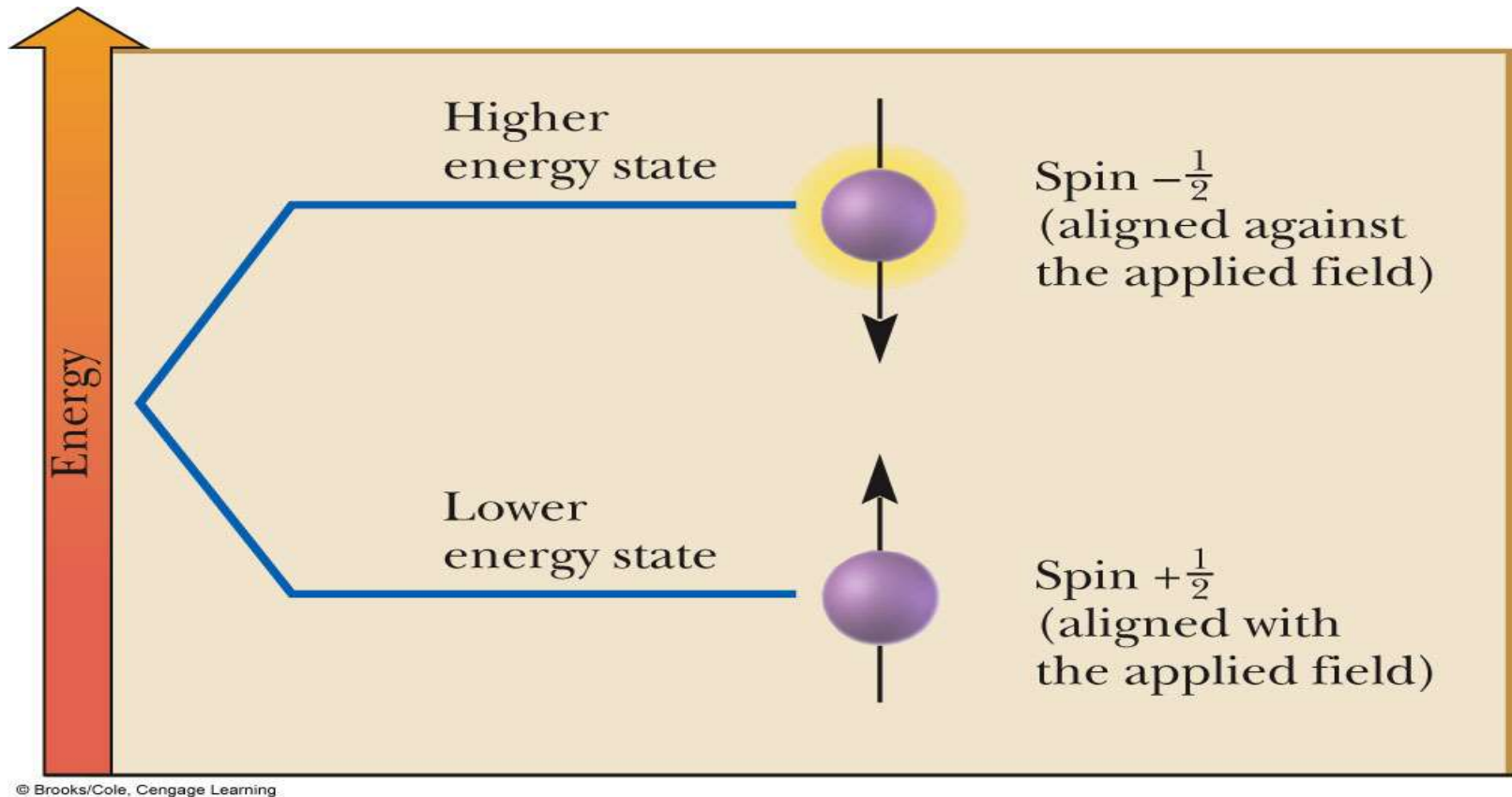


Fig.2. Change in spin state energy separation with increase by applied magnetic field, B_0

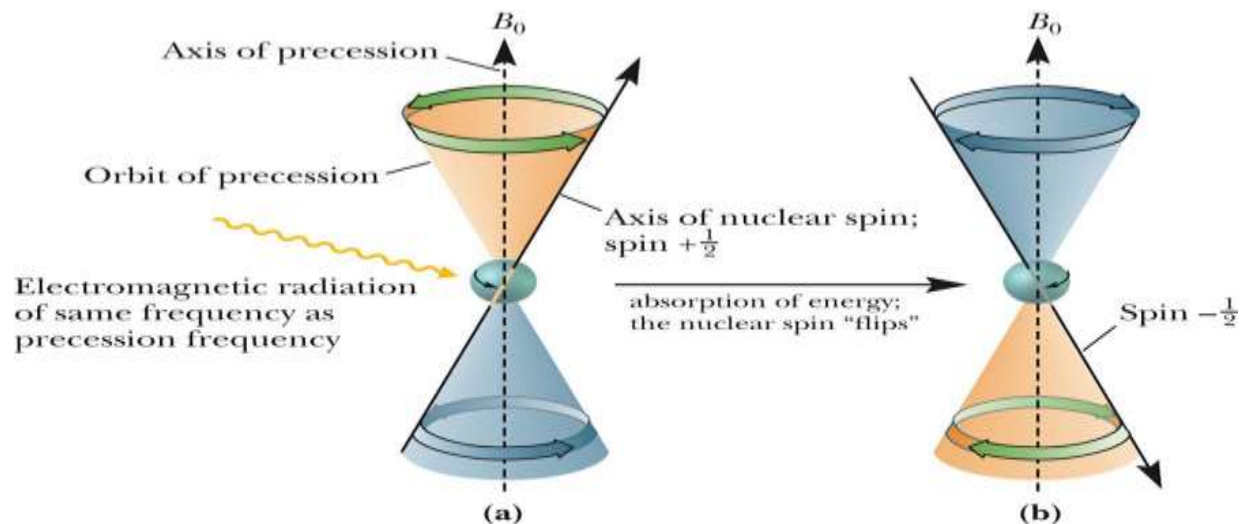
- Stronger the applied magnetic field, greater will be the difference between the spin state, $\Delta E = f(B_0)$ eq. -----1
- As $E_{\text{absorbed}} (\Delta E) = \text{difference in energy levels of two energy states} = h\nu$.
- Each nucleus depending on its charge and mass has its own magnetogyric ratio ' γ ' which is ratio of magnetic dipole moment to angular momentum, and it constant for each nucleus, therefore,
 - $\Delta E = f(\gamma B_0) = h\nu$ -----2
 - $\Delta E = \gamma(h/2\pi B_0) = h\nu$ -----3
 - $\nu = (\gamma/2\pi) B_0$ -----4
- Here, ν is frequency of electromagnetic radiation.

- **Precessional frequency:-** May be defined as revolutions per second made by the magnetic moment vector of the nucleus around the external magnetic field B_0 & Alternatively precessional frequency of spinning bar magnet may be defined as equal to the frequency of EMR in megacycles per second necessary to induce a transition from one spin state to another spin state.
- The energy required to bring transition in NMR ($\Delta E = h\nu$) or to flip proton depends upon the strength of external magnetic field.
- So, stronger the field greater will be the tendency of nucleus magnet to remain lined up with it and higher will be the frequency of radiation needed to flip proton to higher energy state.

Nuclear Magnetic Resonance

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- NMR is phenomenon which occurs when those nuclei which are aligned with applied magnetic field are induce to absorb energy and change their spin state to opposite state, it is called as resonance.
- Fig. 3. shows



Brown/Foote/Iverson
Organic Chemistry 5e
Figure 13.03

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- The proton (Nucleus) +ve charge particle generates an magnetic field on its axis with same frequency as that of applied magnetic field, is called precessional frequency.
- The frequency of precession (proton) is directly proportional to the strength of applied magnetic field.
- If the precessing nucleus is irradiated with electromagnetic radiation of the same frequency as that frequency of precession nucleus, then
 - ▣ The two frequencies couple
 - ▣ Energy is absorbed
 - ▣ The nuclear spin is flipped from spin state $+1/2$ (with the applied field) to $-1/2$ (against the applied field).

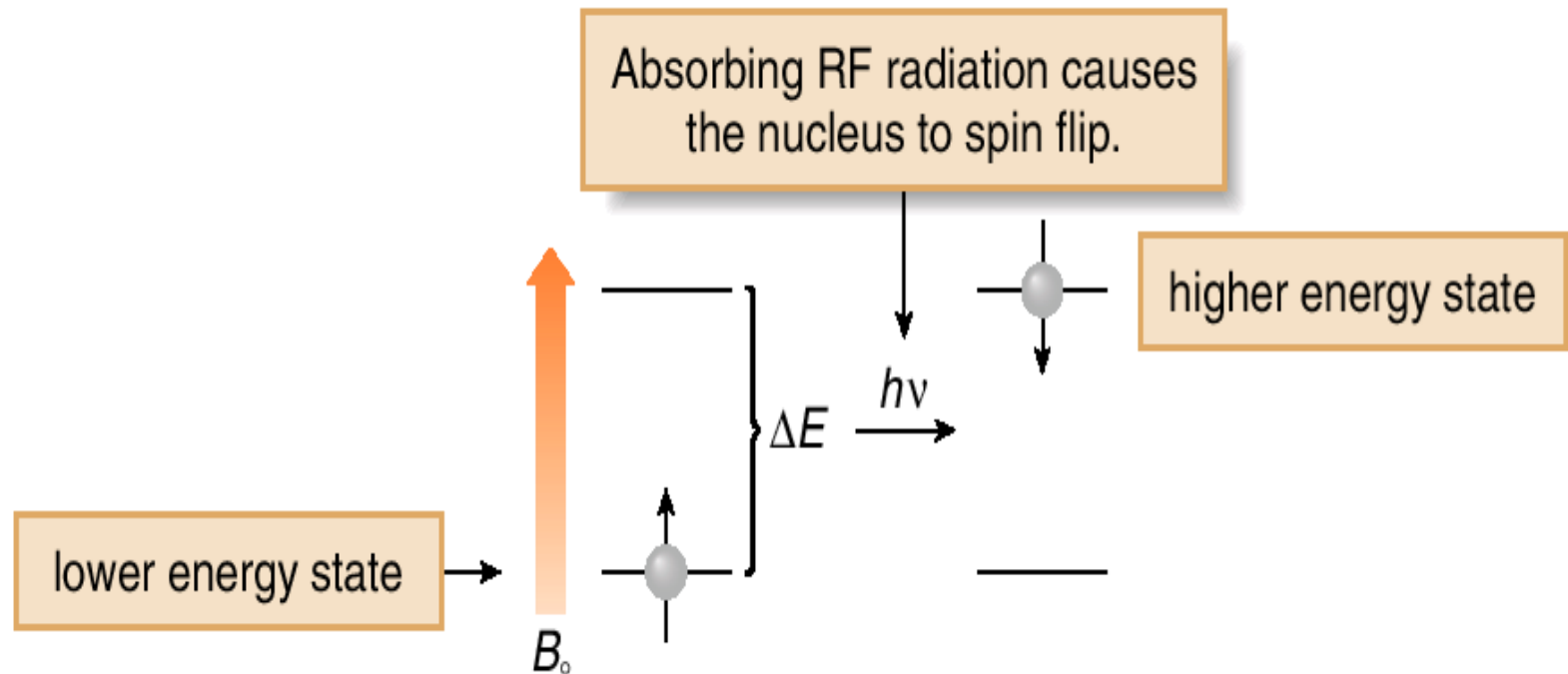


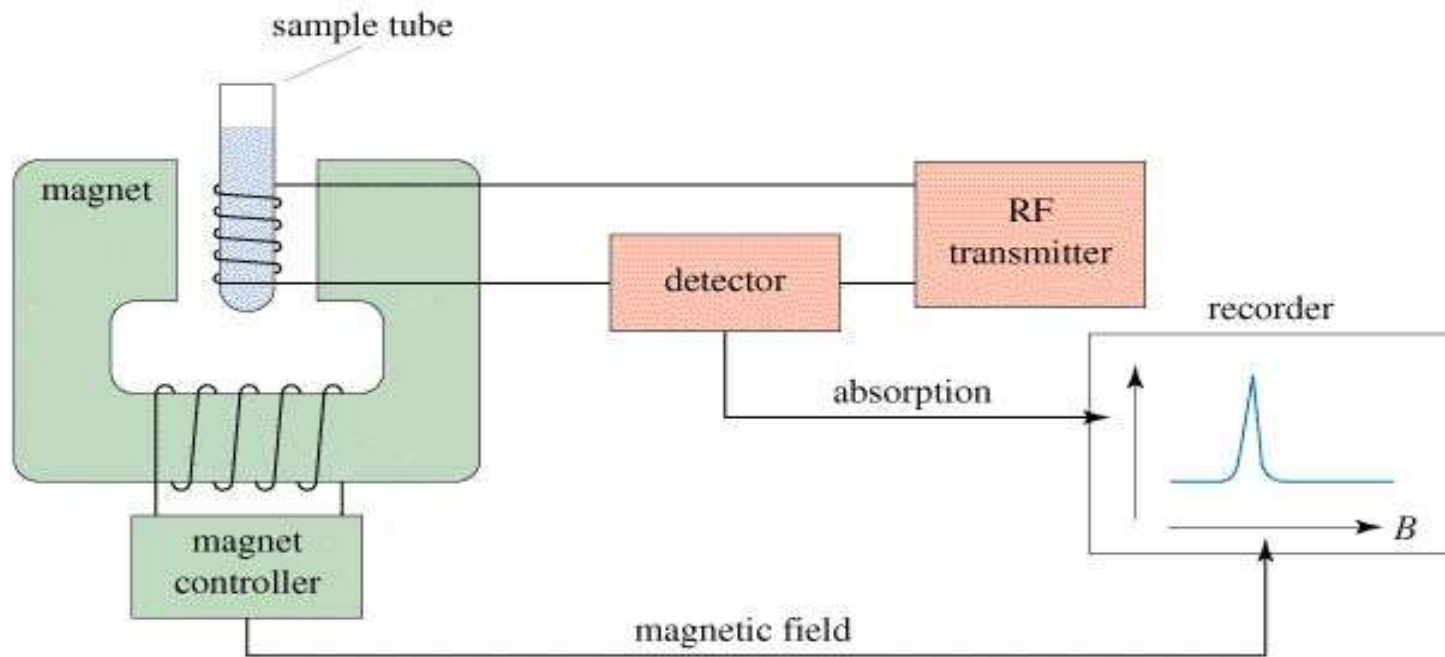
Fig.4. A nucleus is in resonance when it absorbs RF radiation and spin flips to higher energy states.

Instrumentation

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- The NMR spectrophotometer consists of following components;
 - A magnet
 - Sample and sample holder
 - Radiofrequency generator
 - Detector
 - Recorder
- In NMR spectrophotometer, the sample is dissolved in CDCl_3 and placed in magnetic field. Then radiofrequency generator irradiates the sample with short pulse of radiation causing resonance. When nuclei fall back to their low energy state, detector measures the energy released and spectrum is recorded. The superconducting magnet in modern NMR spectrometers have coils that are cooled in liquid helium and conduct electricity with no resistance.

- A line diagram of NMR spectrophotometer along with its components are as follows;



- **Magnet:-** A strong magnet provides stable and homogenous field. The magnet size is 15 inches in diameter and capable of producing strong fields up to 23,500 gauss for 100MHz.
- **Sample and sample holder:-** A 1 – 30 mg sample is used in the form of dilute solution (2 – 10%) and solvent doesn't contain hydrogen of its own ions. The sample holder is glass tube about 5mm in diameter and 15 – 20 cm in length.
- **Radiofrequency oscillator:-** The RF oscillator is installed perpendicular to magnetic field and transmits radio waves of some mixed frequency such as 60, 100, 220, 300 MHz. a sweep generator is installed to supply dc current to sec. magnet.

- **RF detector or Receiver**:- It is installed perpendicular to both magnetic field and the oscillator coil and is tuned to the same frequency as transmitter. When precession frequency is match with RF the nuclei induces (emf) in detector coil and this signal is amplified and sent to recorder.
- **Recorder** :- The recorder gives a spectrum as a plot of strength resonance signal on Y axis & strength of magnetic field on X axis. The strength of resonance signal is directly proportional to number of nuclei resonating at that particular field strength.

Working of instrument

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- Sample placed in long cylindrical glass tube especially made for NMR.
- Dissolved the sample in proton free solvent like CDCl_3 or CCl_4 and add small amount of TMS as internal reference.
- Placed the sample in gap between two magnetic poles where coil is attached to a specific RF generator (e.g. 60 MHz).
- This coil supply EMR energy required to change spin orientation of proton.
- Then detector coil detects radiofrequency signal when resonance occurs.
- As magnetic field strength increase, the precessional frequency of all protons increase and when this frequency proton reaches

60 MHz, the resonance occurs.

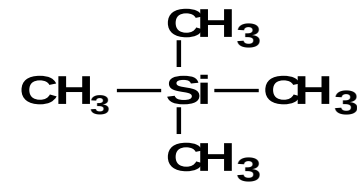
- So, as magnetic field increase linearly the recorder pen travels from left to right, thus protons which achieve resonance faster i.e. (Deshielded) appears on left side (downfield), where as those protons (Shielded) appears on right side (upfield) of chart in the form of peaks.
- TMS is shown at a peak value of $\delta = 0$ ppm.

Solvents

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- The solvent used for dissolving sample should have following properties;
 - Should not contain proton,
 - Inexpensive
 - Low boiling point and non polar in nature.
- Generally deuterated chloroform CDCl_3 is used as solvent.
- If sample is soluble in polar solvent, then deuterium oxide (D_2O), DMSO, CCl_4 , CS_2 , Cf_3 , COOH are used as solvent.

Internal Standard



Tetramethylsilane (TMS)

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- But TMS (Tetra methyl silane) is most commonly used as IS for measuring the position of ^1H , ^{13}C and ^{29}Si in NMR spectroscopy. Due to following reasons;
 - It is chemically inert and miscible with a large range of solvents.
 - Its twelve protons are all magnetically equivalent.
 - Its protons are highly shielded and gives a strong peak even small quantity.
 - It is less electronegative than carbon.
 - It is highly volatile and can be easily removed to get back sample.
 - It does not take part in intermolecular associations with sample.

- Its resonance position is far away from absorptions due to protons in most organic molecules, thus signals of TMS = 0.

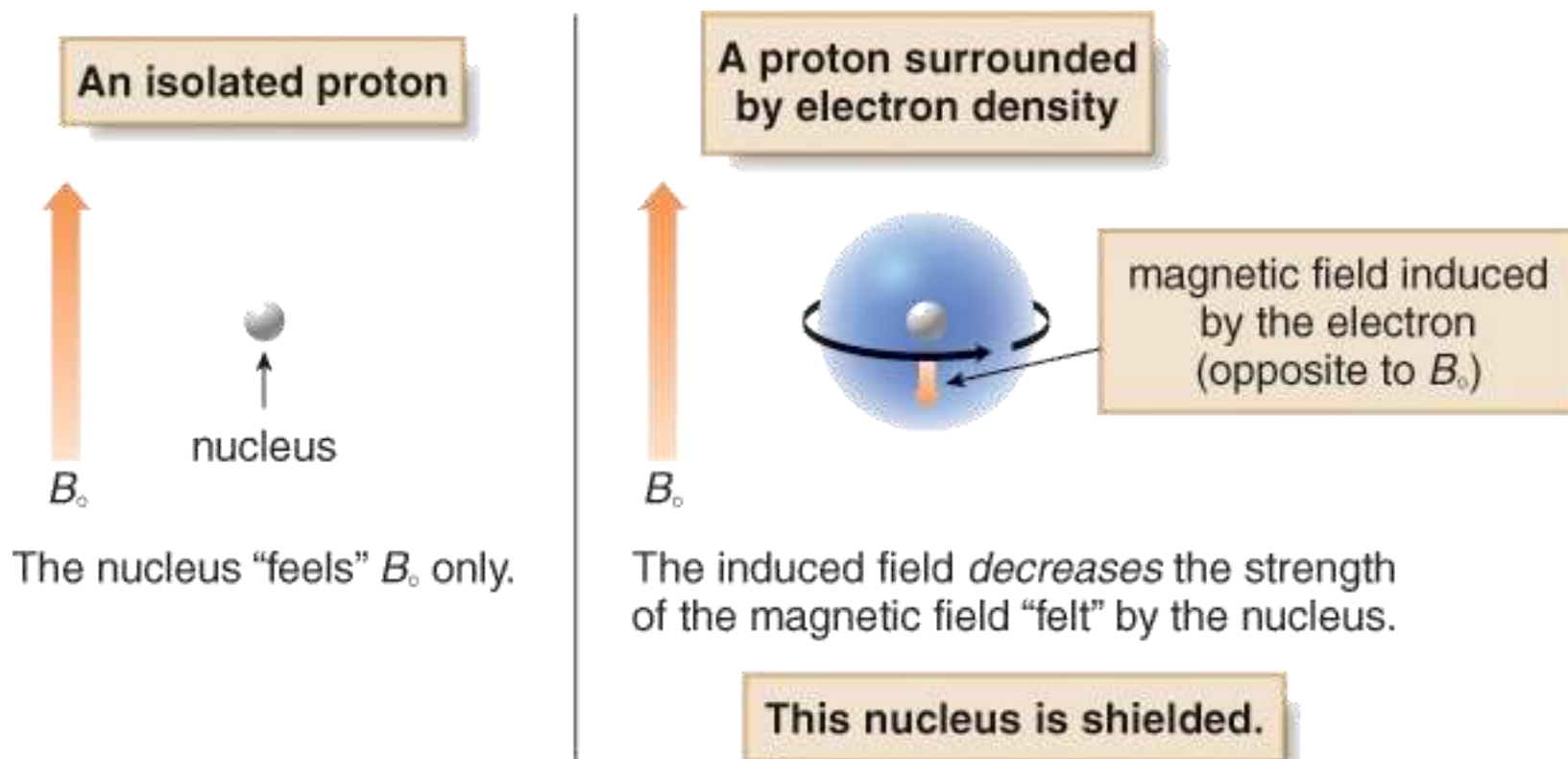
Chemical Shift (Position of Signals)

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- The utility of NMR is that all protons do not show resonance at same frequency bcoz, it is surrounded by particular no. of valence electrons which vary from atom to atom so, they exist in slightly different electronic environment from one another.
- Position of signals in spectrum help us to know nature of protons i.e. aromatic, aliphatic, acetylenic, vinylic, adjacent to electron releasing or withdrawing grp. Thus they absorb at different field strength.
- When molecule placed in magnetic field, so its surrounding electron circulate & generates counter field which opposes the applied magnetic field on proton so that, field feels by proton is reduced and that proton called as the **Shielded** proton.

- Rotation of electrons (π) to nearby nuclei generate field that can either oppose or strong the field on proton.
- If magnetic field is oppose applied magnetic field on proton, that proton said shielded proton and if field is strong the applied field then, proton feels high magnetic field strength and such proton called as **Deshielded** proton.
- So, shielded proton shifts absorption signal to right side (upfield) and deshielded proton shifts absorption signal to left side (down field) of spectrum.
- So, electric environment surrounding proton tells us where proton shows absorption in spectrum.

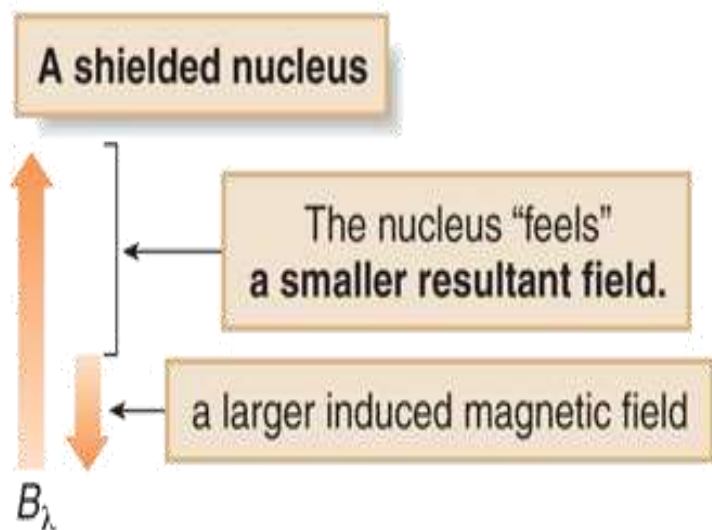
- Such shifting in position of NMR absorption signals which arise due to the shielding or deshielding of proton by surrounding electrons are called as **Chemical shift**.



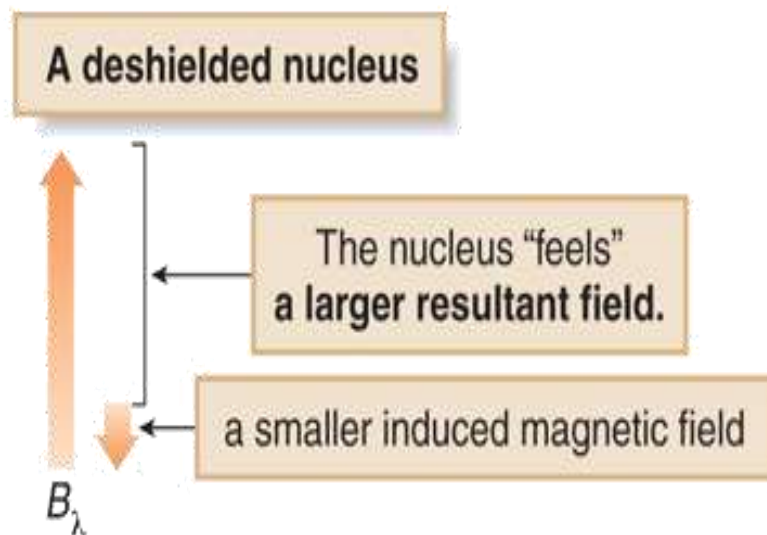
Shielding or Deshielding Protons In Molecule

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- As we have seen, depending on electronic environment protons in molecules are shielded or deshielded by different amounts.



- As the electron density around the nucleus increases, the nucleus feels a smaller resultant magnetic field, so a lower frequency is needed to achieve resonance.
- **The absorption shifts upfield.**

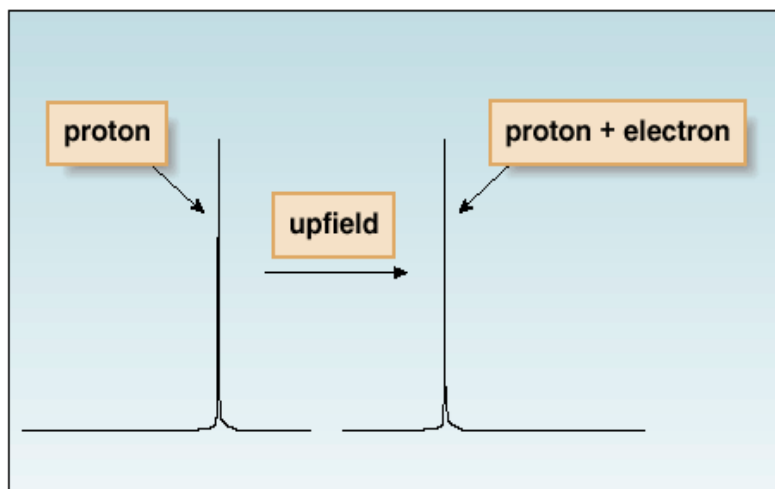


- As the electron density around the nucleus decreases, the nucleus feels a larger resultant magnetic field, so a higher frequency is needed to achieve resonance.
- **The absorption shifts downfield.**

- So, shielded proton shows absorption of signals to right side and deshielded protons at left side of spectrum.

a. Shielding effects

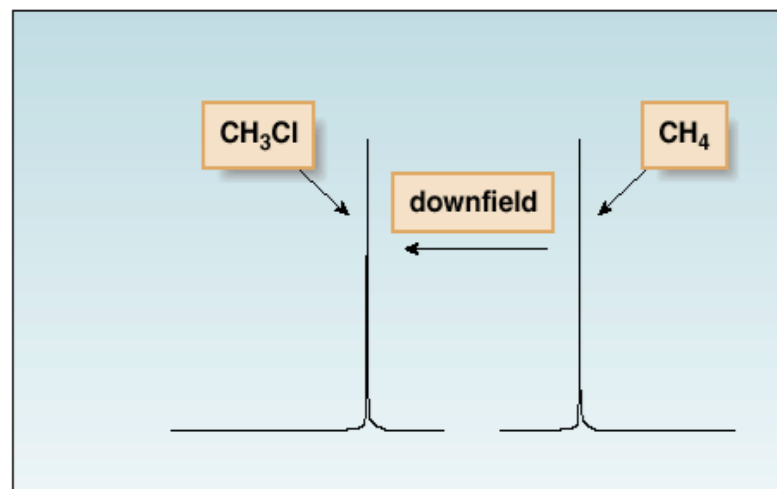
- An electron shields the nucleus.
- The absorption shifts *upfield*.



Increasing chemical shift
Increasing ν

b. Deshielding effects

- Decreased electron density deshields a nucleus.
- The absorption shifts *downfield*.



Increasing chemical shift
Increasing ν

- For measuring chemical shifts of various protons in a molecule, the signal of TMS is taken as reference, due to low electronegativity of Si atom and shielding of equivalent protons gives 1 NMR signal is greater than most of organic compounds.
- The nmr signal for particular protons will appear at different field strength compared to signal from TMS.
- ***This difference in the absorption position of the proton with respect to TMS signal is called as chemical shift (δ – value).***
- In majority of organic compound, protons resonate at lower field than protons of TMS coz, its signals appears at the extreme right side of spectrum, thus assigning delta (δ) value for TMS equal to zero (0).

- Any proton or set of protons which absorb at lower field than TMS is given a positive value for (δ).
- The value of δ for a substance w.r.p to TMS can be obtained by
$$\text{Chemical shift, ppm } \delta = \frac{\text{Shift downfield from TMS (in Hz)}}{\text{Spectrometer frequency (in MHz)}}$$
- The value of chemical shift, δ expressed in ppm and their value is between 0 to 10 in the δ scale.
- In the τ scale, signal for standard reference, TMS taken as 10 ppm.
- The two scales are related by the expression

$$\tau = 10 - \delta$$

Chemical shift depends upon following parameters:

- **Electro negativity of nearby atoms**
- **Hybridization of adjacent atoms**
- **Diamagnetic effects from adjacent pi bonds**

□ Electro negativity of nearby atoms

CH₃ - X	Electronegativity of X	Chemical Shift (δ)
CH₃ F	4.0	4.26
CH₃ OH	3.5	3.47
CH₃ Cl	3.1	3.05
CH₃ Br	2.8	2.68
CH₃ I	2.5	2.16
(CH₃)₄ C	2.1	0.86
(CH₃)₄ Si	1.8	0.00

□ Hybridization of adjacent atoms

Type of Hydrogen (R = alkyl)	Name of Hydrogen	Chemical Shift (δ)
RCH_3 , R_2CH_2 , R_3CH	Alkyl	0.8 - 1.7
$\text{R}_2\text{C}=\text{C}(\text{R})\text{CHR}_2$	Allylic	1.6 - 2.6
$\text{RC}\equiv\text{CH}$	Acetylenic	2.0 - 3.0
$\text{R}_2\text{C}=\text{CHR}$, $\text{R}_2\text{C}=\text{CH}_2$	Vinylic	4.6 - 5.7
RCHO	Aldehydic	9.5-10.1

□ Diamagnetic effects from adjacent pi bonds

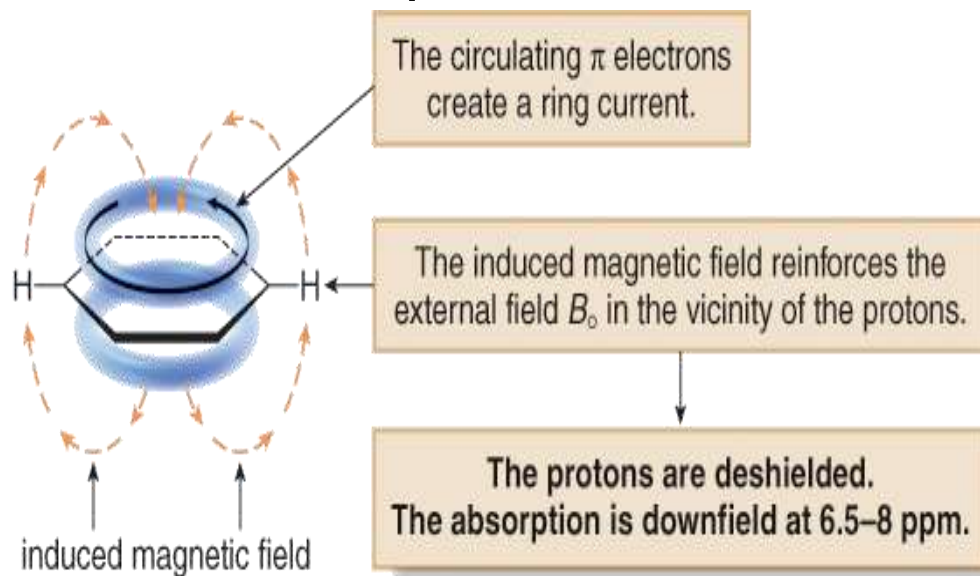
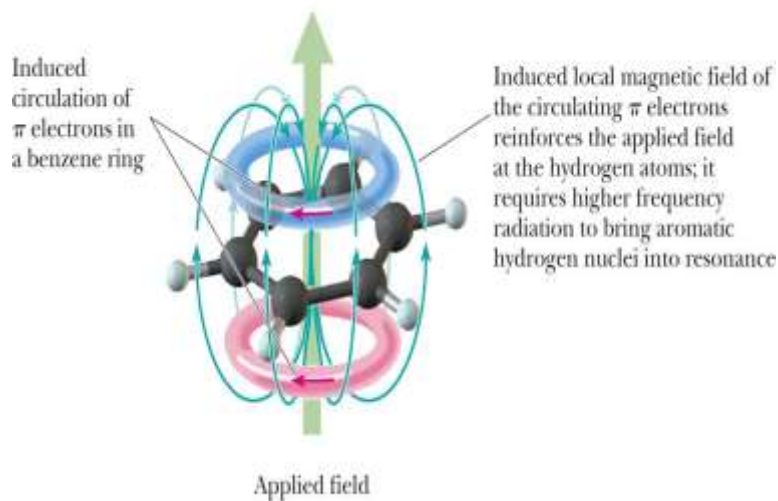
- A carbon-carbon triple bond shields an acetylenic hydrogen and shifts its signal to lower frequency (to the right) to a smaller δ value.
- A carbon-carbon double bond deshields vinylic hydrogens and shifts their signal to higher frequency (to the left) to a larger δ value.

Type of H	Name	Chemical Shift (δ)
RCH_3	Alkyl	0.8- 1.0
$\text{RC}\equiv\text{CH}$	Acetylenic	2.0 - 3.0
$\text{R}_2\text{C}=\text{CH}_2$	Vinylic	4.6 - 5.7

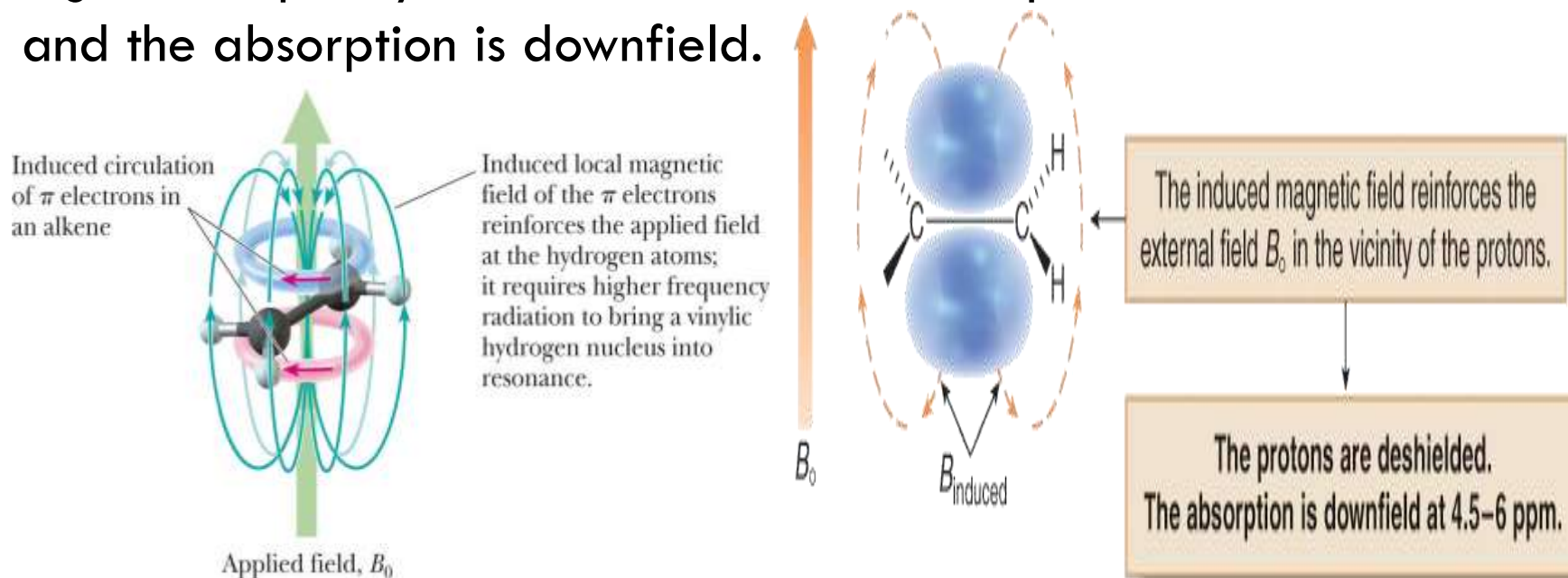
Chemical Shift Values (Benzene)

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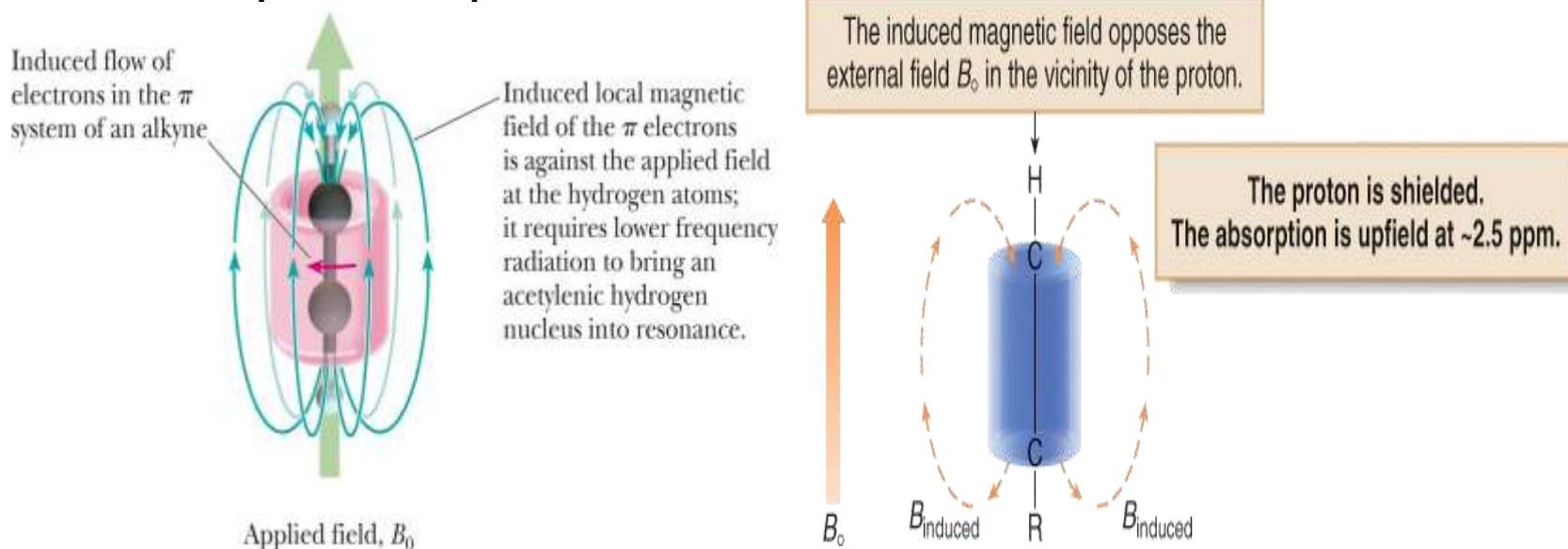
- ❑ In a magnetic field, the six π electrons in benzene circulate around the ring creating a ring current.
- ❑ The magnetic field induced by these moving electrons reinforces the applied magnetic field in the vicinity of the protons.
- ❑ The protons thus feel a stronger magnetic field and a higher frequency is needed for resonance. Thus they are deshielded and absorb downfield.



- ❑ In a magnetic field, the loosely held π electrons of the double bond create a magnetic field that reinforces the applied field in the vicinity of the protons.
- ❑ The protons now feel a stronger magnetic field, and require a higher frequency for resonance. Thus the protons are deshielded and the absorption is downfield.



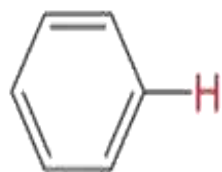
- In a magnetic field, the π electrons of a carbon-carbon triple bond are induced to circulate, but in this case the induced magnetic field opposes the applied magnetic field.
- Thus, the proton feels a weaker magnetic field, so a lower frequency is needed for resonance. The nucleus is shielded and the absorption is upfield.



Proton type

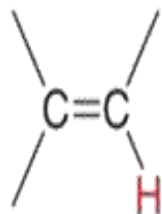
Effect

Chemical shift (ppm)



highly deshielded

6.5–8



deshielded

4.5–6



shielded

~2.5

Factors affecting chemical shift

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Following are the factors which influence the chemical shift;

- **Inductive effect**
- **Van der Waal's deshielding**
- **Anisotropic effect**
- **Hydrogen bonding**

Spin-Spin Splitting in NMR Spectrum

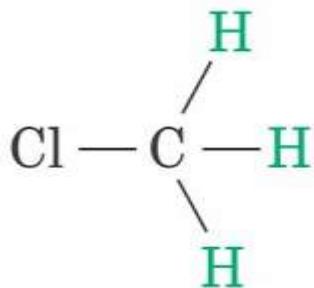
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- Peaks are often split into multiple peaks due to magnetic interactions between **nonequivalent protons on adjacent carbons**, The process is called **Spin-spin Splitting/Coupling**.
- **Spin-spin splitting**: It occurs only between nonequivalent protons on the same carbon or adjacent carbons.
- **Peak**: The units into which an NMR signal is split; doublet, triplet, quartet, multiplet, etc.
- **Signal splitting**: Splitting of an NMR signal into a set of peaks by the influence of neighboring nonequivalent hydrogens.
- **Signal coupling**: An interaction in which the nuclear spins of adjacent atoms influence each other and lead to the splitting of NMR signals.

Equivalent & Nonequivalent Protons

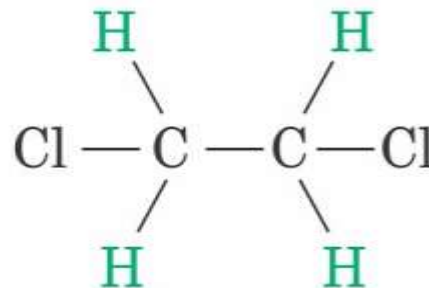
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- Equivalent protons are like this;



Three C–H protons are chemically equivalent; no splitting occurs.

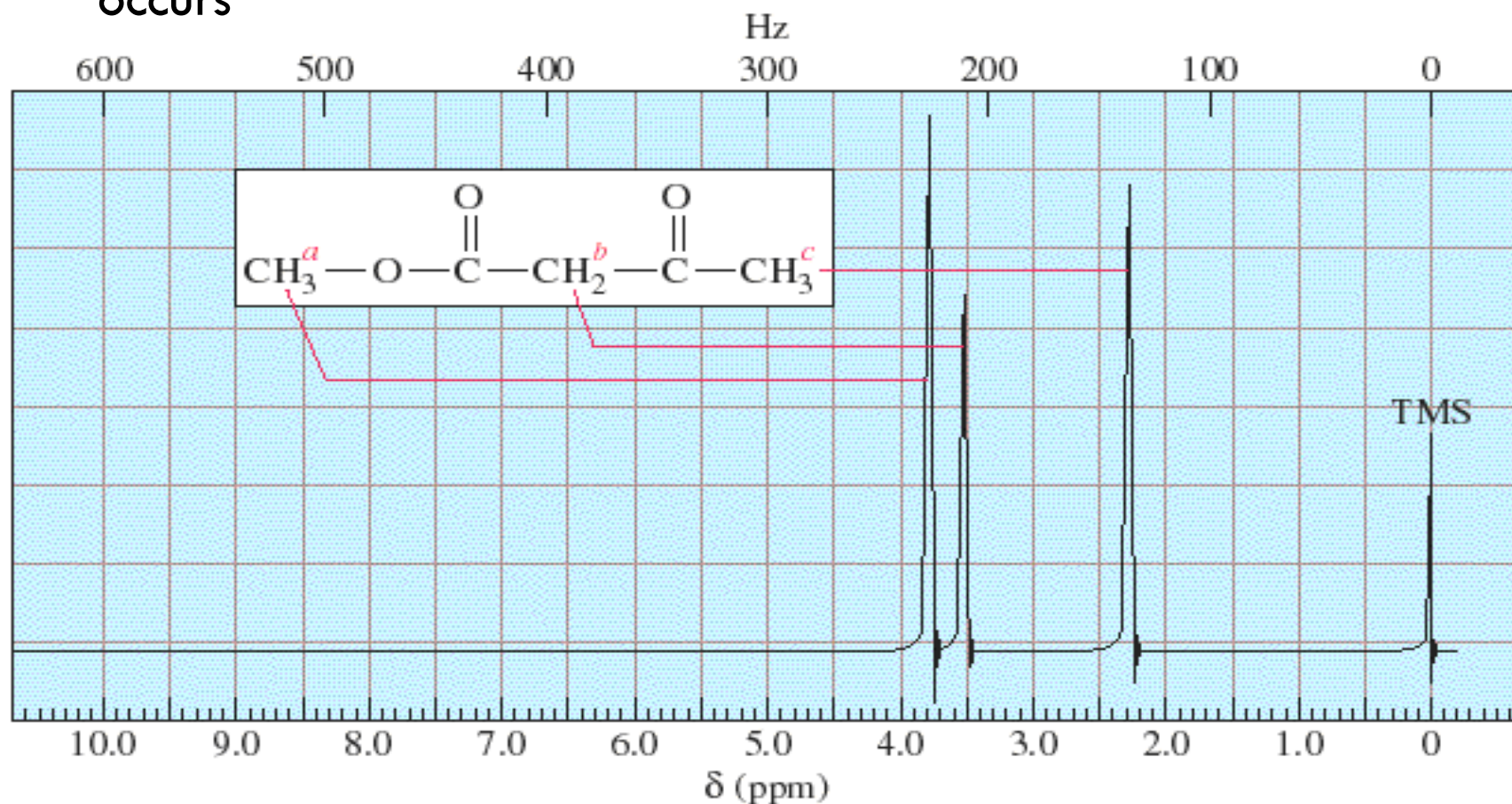
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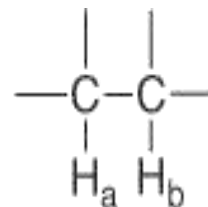
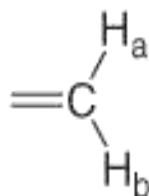
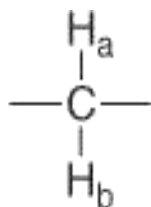
Four C–H protons are chemically equivalent; no splitting occurs.

- These equivalent protons do not split each other.
- Protons bonded to the same carbon will split each other only if they are not equivalent.
- Equivalent protons have the same chemical shift.

- Equivalent protons have same chemical shift without splitting occurs



- Nonequivalent protons are like this;

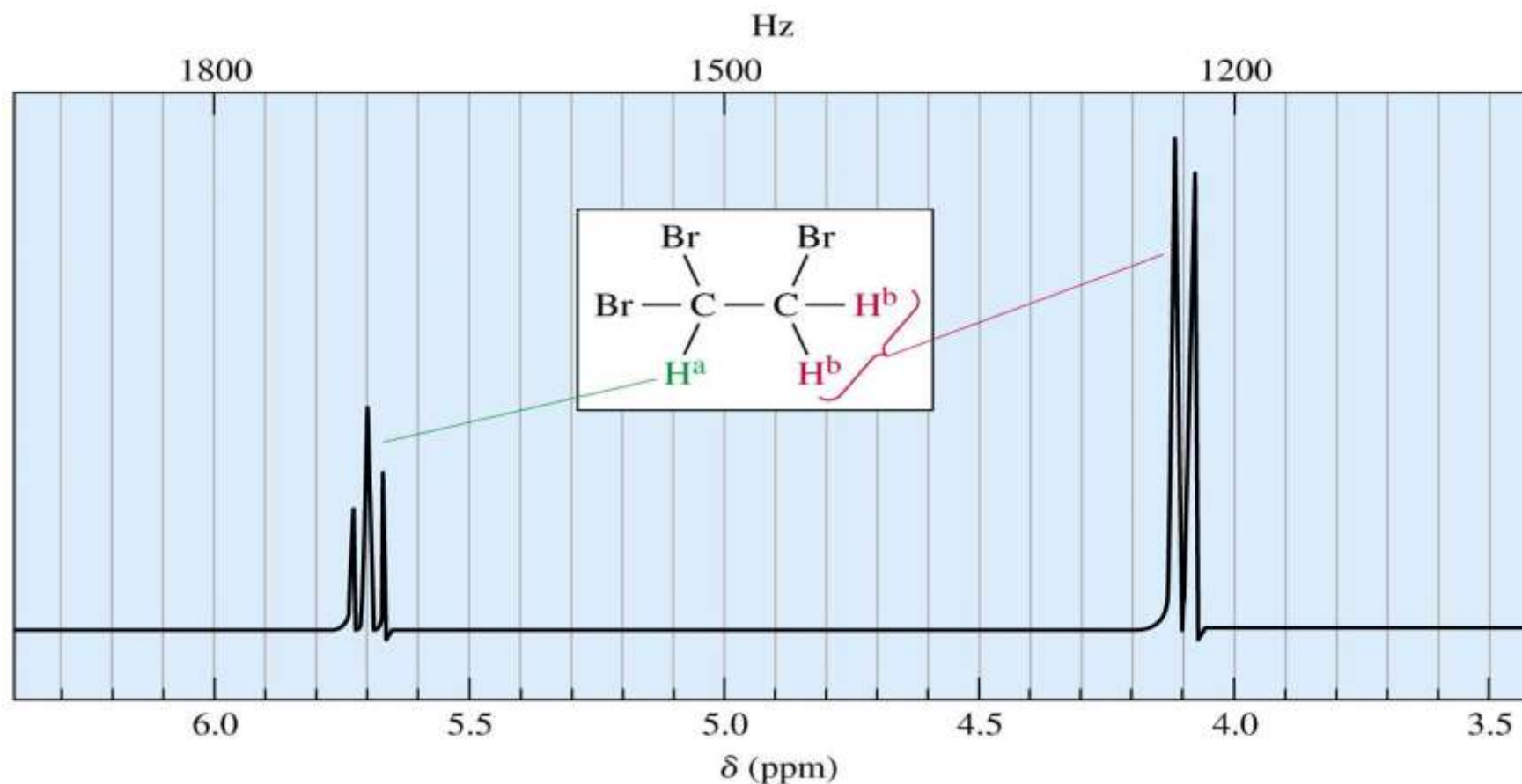


H_a and H_b are on the **same** carbon.

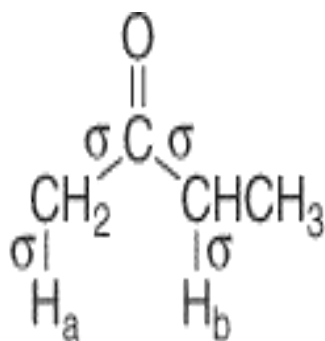
H_a and H_b are on **adjacent** carbons.

- If H_a and H_b are not equivalent, the splitting is observed.
- Nonequivalent protons on adjacent carbons have magnetic fields that may align with or oppose the external field.
- This magnetic coupling causes the proton to absorb slightly downfield when the external field is reinforced and slightly upfield when the external field is opposed.
- All possibilities exist, so signal is split.

- Non equivalent protons have different chemical shift with splitting occurs



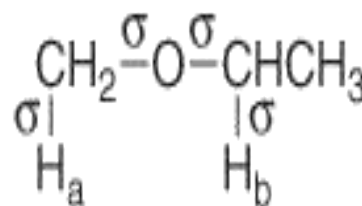
- Splitting is not generally observed between protons separated by more than three σ bonds.



2-butanone

H_a and H_b are separated by four σ bonds.

no splitting between H_a and H_b



ethyl methyl ether

H_a and H_b are separated by four σ bonds.

no splitting between H_a and H_b

- **Peak:** The units into which an NMR signal is split; **doublet**, **triplet**, **quartet**, **multiplet**, etc.
- So now, let us consider how the **doublet signal** due to the CH_2 group on $\text{BrCH}_2\text{CHBr}_2$ occurs:
 - When placed in an applied field, (B_0), the adjacent proton (CHBr_2) can be aligned with ($\uparrow$) or against ($\downarrow$) B_0 .
 - Thus, the absorbing CH_2 protons feel two slightly different magnetic fields—one slightly larger than B_0 , and one slightly smaller than B_0 .
 - Since the absorbing protons feel two different magnetic fields, they absorb at two different frequencies in the NMR spectrum, thus splitting a single absorption into a doublet, where the two peaks of the doublet have equal intensity.

How doublet signal arises

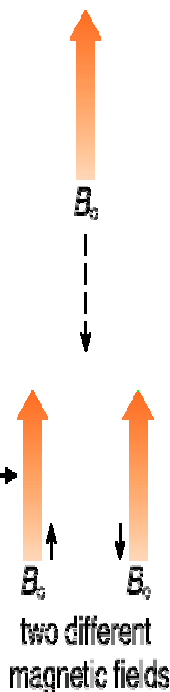
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How a doublet arises

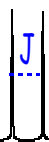
With no adjacent H's:
The absorbing H's feel only one magnetic field.



With one adjacent H:
The absorbing H's feel two different fields,
so they absorb at two different frequencies.

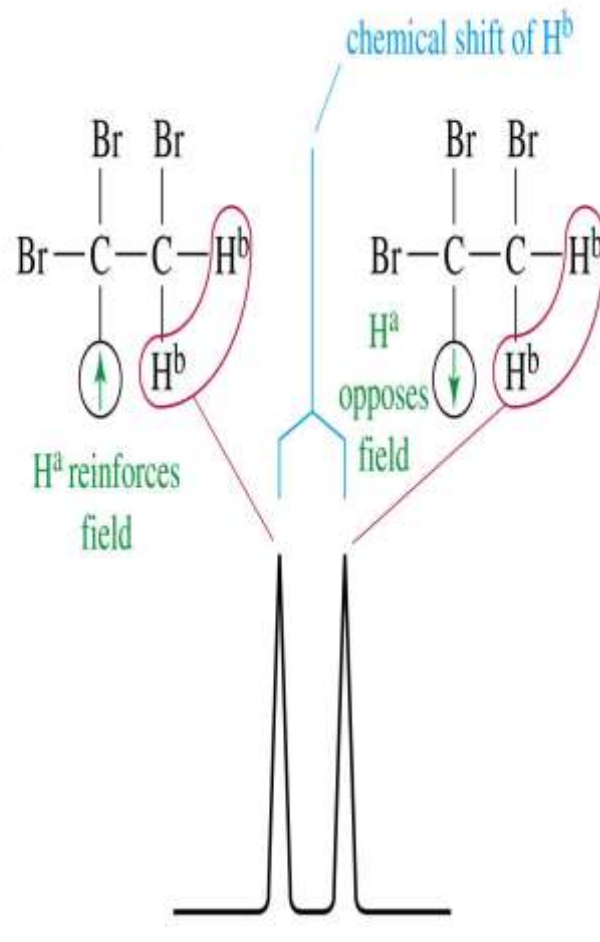


The NMR signal is
a **single peak**.



The NMR signal is split
into a **doublet**.

1:1



- One adjacent proton splits an NMR signal into a doublet.

- Let us now consider how a **triplet** arises:

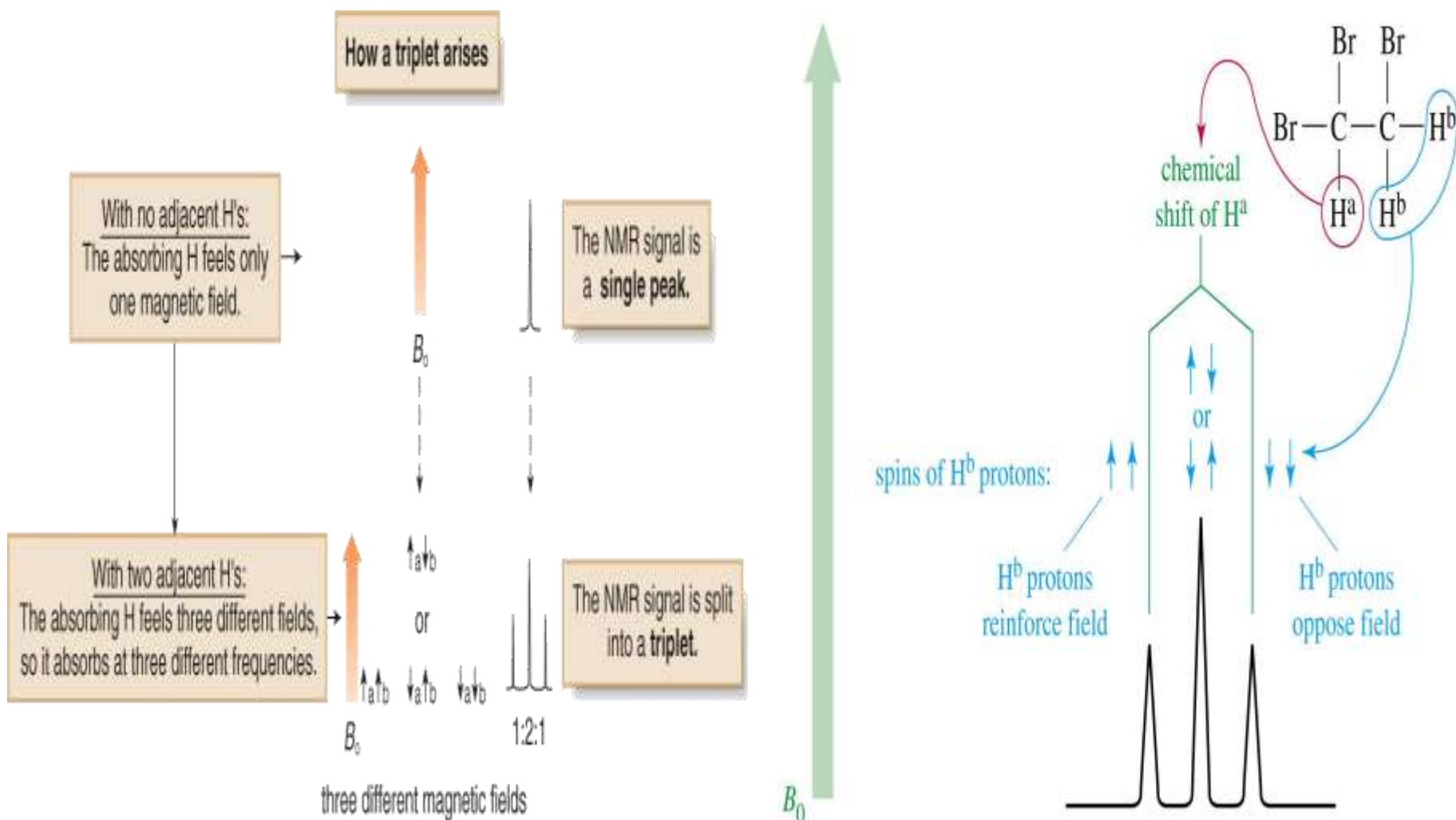


- When placed in an applied magnetic field (B_0), the adjacent protons H_a and H_b can each be aligned with ($\uparrow$) or against ($\downarrow$) B_0 .
- Thus, the absorbing proton feels three slightly different magnetic fields—one slightly larger than B_0 ($\uparrow_a \uparrow_b$), one slightly smaller than B_0 ($\downarrow_a \downarrow_b$) and one the same strength as B_0 ($\uparrow_a \downarrow_b$).

- Because the absorbing proton feels three different magnetic fields, it absorbs at three different frequencies in the NMR spectrum, thus splitting a single absorption into a **triplet**.
- Because there are two different ways to align one proton with B_0 , and one proton against B_0 —that is, $\uparrow a \downarrow b$ and $\downarrow a \uparrow b$ —the middle peak of the triplet is twice as intense as the two outer peaks, making the ratio of the areas under the three peaks 1:2:1.
- Two adjacent protons split an NMR signal into a triplet.
- When two protons split each other, they are said to be coupled.
- The spacing between peaks in a split NMR signal, measured by the J value, is equal for coupled protons.

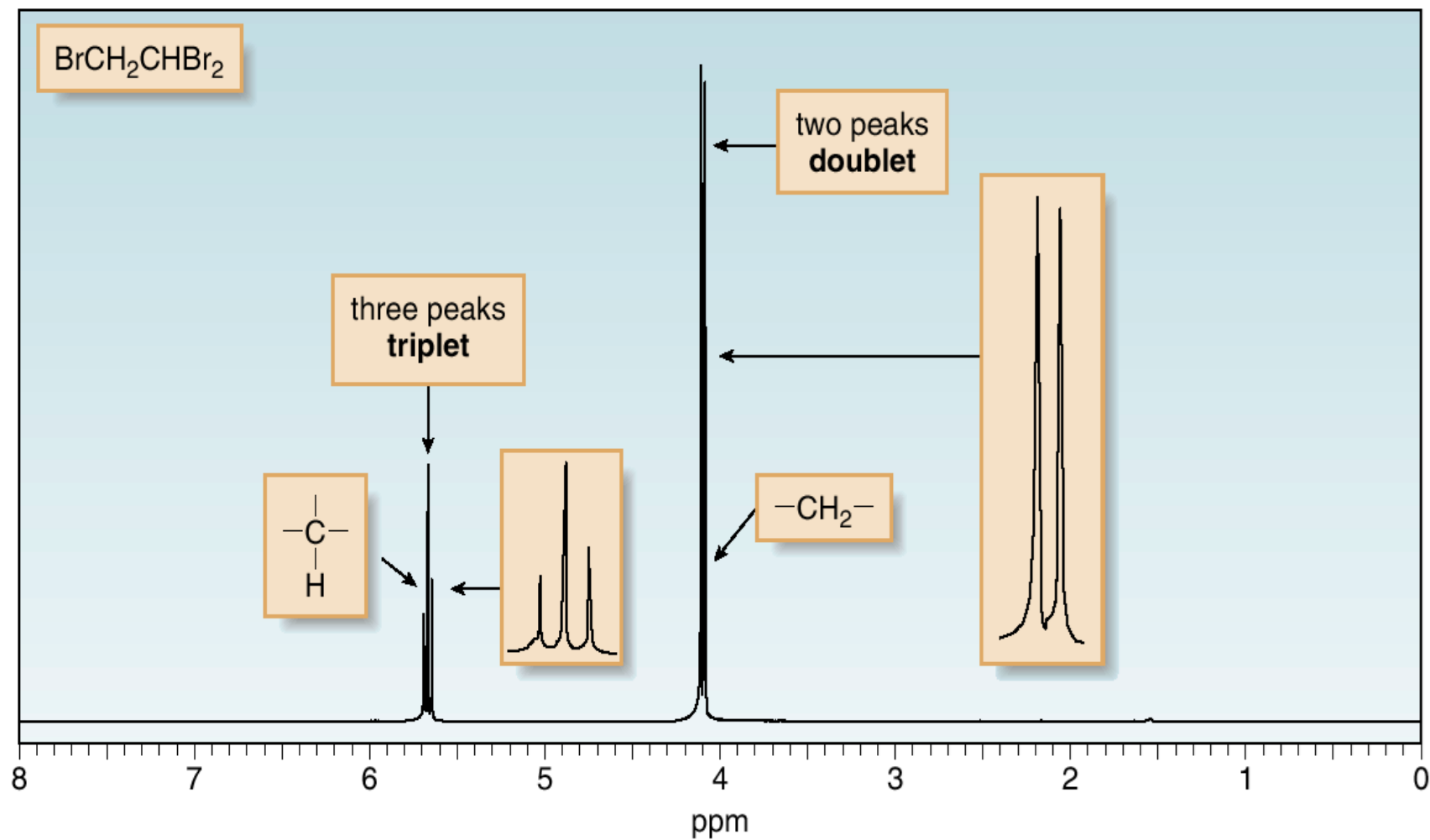
How triplet signal arises

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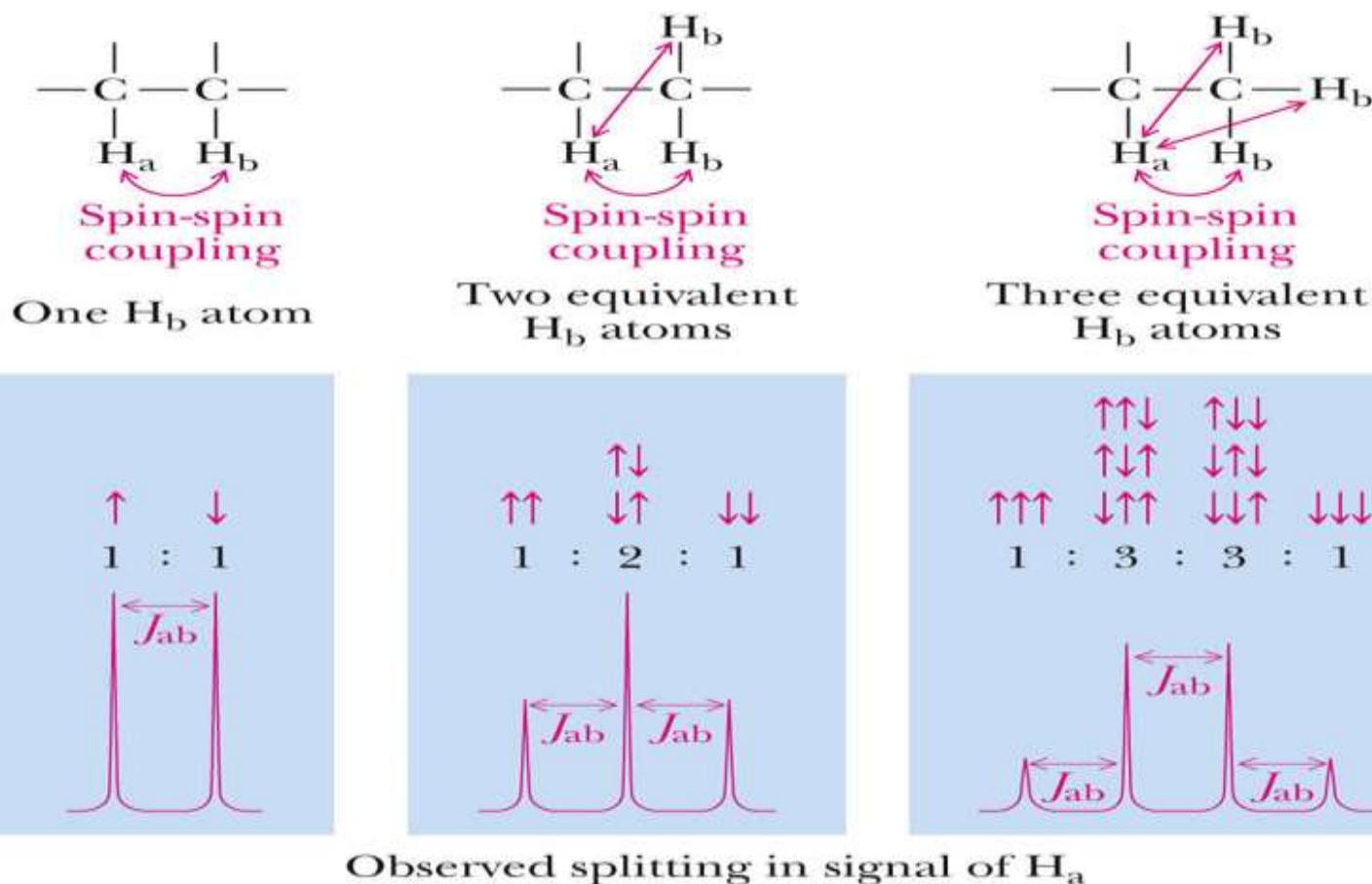
Doublet and triplet signals

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How quartet signal arises

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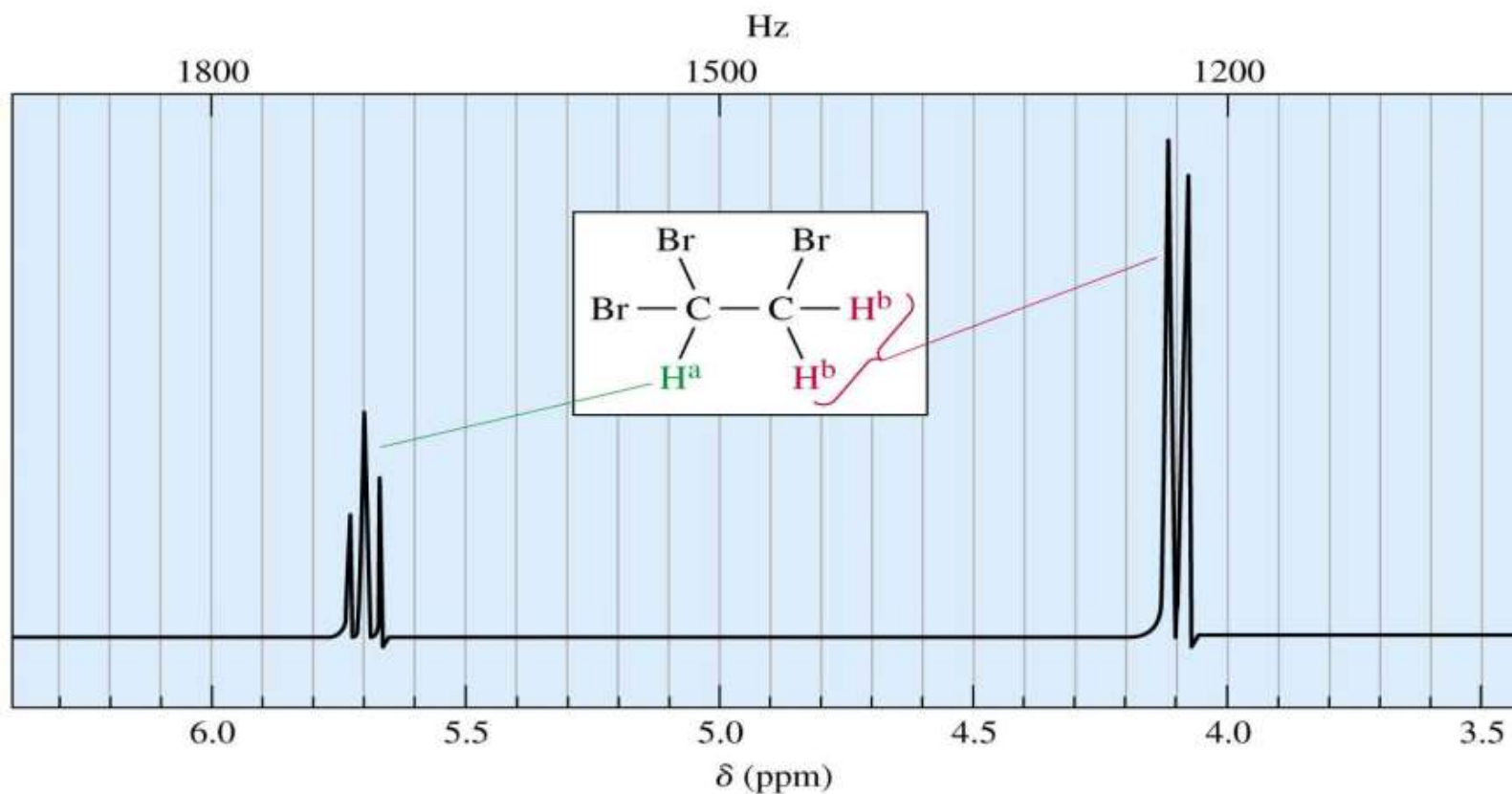


Signal Splitting: The $(n + 1)$ Rule

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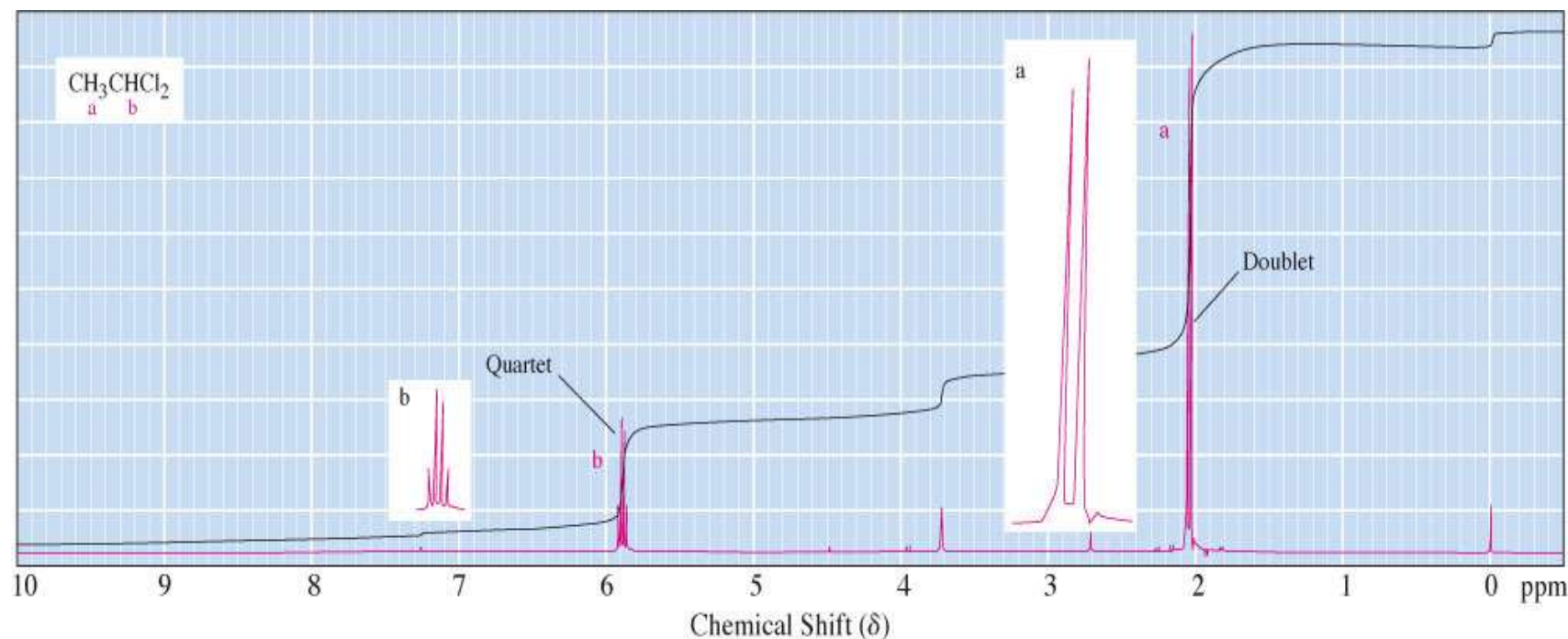
- In a NMR spectrum all equivalent protons do not appear as a signal peak, e.g. **1,1,2 - tribromoethane** which has two types of equivalent protons, thus it shows two peaks in NMR spectrum.
- But the actual spectrum consists of two peaks but subdivided into 3 and 2 sub peaks or splitting one for (CH) and two for (CH₂Br) protons respectively.
- This phenomenon of splitting of equivalent protons into $(n + 1)$ rule, where **n** is **the no. of equivalent protons attached to the adjacent carbon to which the protons under consideration is attached**.
- So as per the $(n + 1)$ rule in **1,1,2- tribromoethane**, the (C₁) has two equivalent protons of Methylene on the carbon next to it, therefore, $n = 2$ and hence it will split into $(2+1) = 3$ peaks (Triplet).

- The Methylene protons are having $n = 1$ therefore, it will split $(1+1) = 2$ peaks (Doublet).



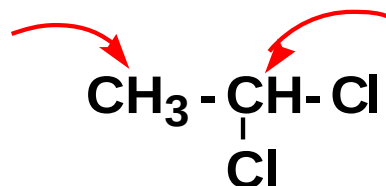
Another example

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For these hydrogens, $n = 1$;
their signal is split into
 $(1 + 1) = 2$ peaks; a doublet



For this hydrogen, $n = 3$;
its signal is split into
 $(3 + 1) = 4$ peaks; a quartet

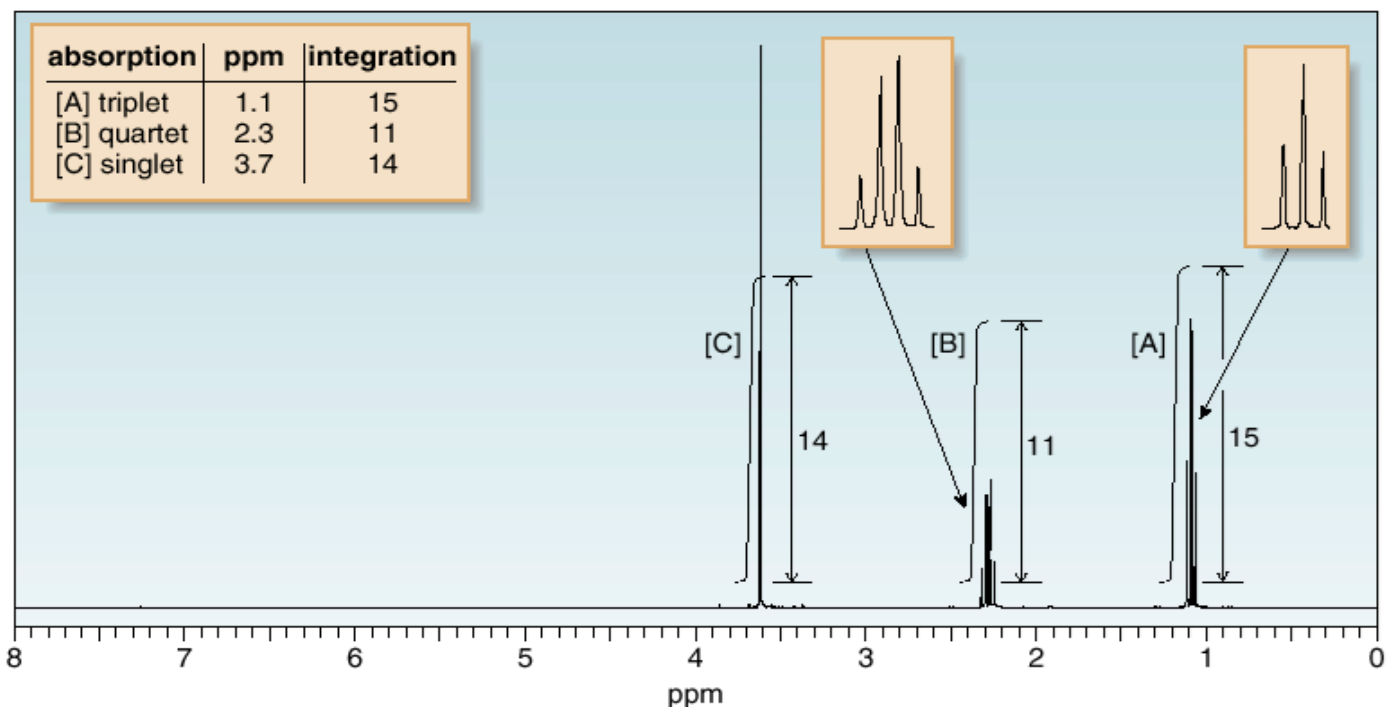
^1H NMR—Structure Determination

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How To

Use ^1H NMR Data to Determine a Structure

Example Using its ^1H NMR spectrum, determine the structure of an unknown compound **X** that has molecular formula $\text{C}_4\text{H}_8\text{O}_2$ and contains a $\text{C}=\text{O}$ absorption in its IR spectrum.



Step [1] Determine the number of different kinds of protons.

- The number of NMR signals equals the number of different types of protons.
- This molecule has three NMR signals ([A], [B], and [C]) and therefore **three** types of protons (H_a , H_b , and H_c).

How To, continued . . .

Step [2] Use the integration data to determine the number of H atoms giving rise to each signal (Section 14.5).

- Total number of integration units: $14 + 11 + 15 = 40$ units
- Total number of protons = 8
- Divide: $40 \text{ units} / 8 \text{ protons} = \mathbf{5 \text{ units per proton}}$
- Then, divide each integration value by this answer (5 units per proton) and round to the nearest whole number.

$$\frac{15}{5} = 3 \text{ H}_a \text{ protons}$$

signal [A]



Three equivalent H's usually means a **CH₃** group.

$$\frac{11}{5} = 2.2 \approx 2 \text{ H}_b \text{ protons}$$

signal [B]



Two equivalent H's usually means a **CH₂** group.

$$\frac{14}{5} = 2.8 \approx 3 \text{ H}_c \text{ protons}$$

signal [C]

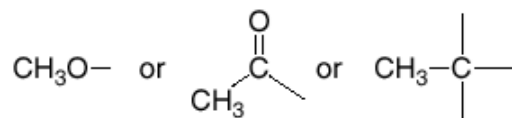


Three equivalent H's usually means a **CH₃** group.

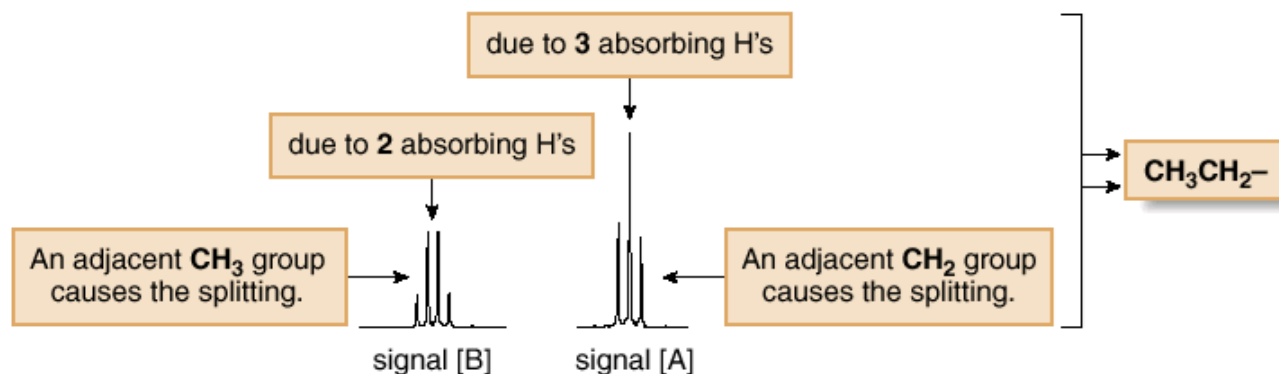
How To, continued . . .

Step [3] Use individual splitting patterns to determine what carbon atoms are bonded to each other.

- Start with the singlets. Signal [C] is due to a CH_3 group with no adjacent nonequivalent H atoms. Possible structures include:



- Because signal [A] is a **triplet**, there must be **2 H's** (CH_2 group) on the adjacent carbon.
- Because signal [B] is a **quartet**, there must be **3 H's** (CH_3 group) on the adjacent carbon.
- This information suggests that **X** has an **ethyl** group $\text{---} \rightarrow \text{CH}_3\text{CH}_2-$.

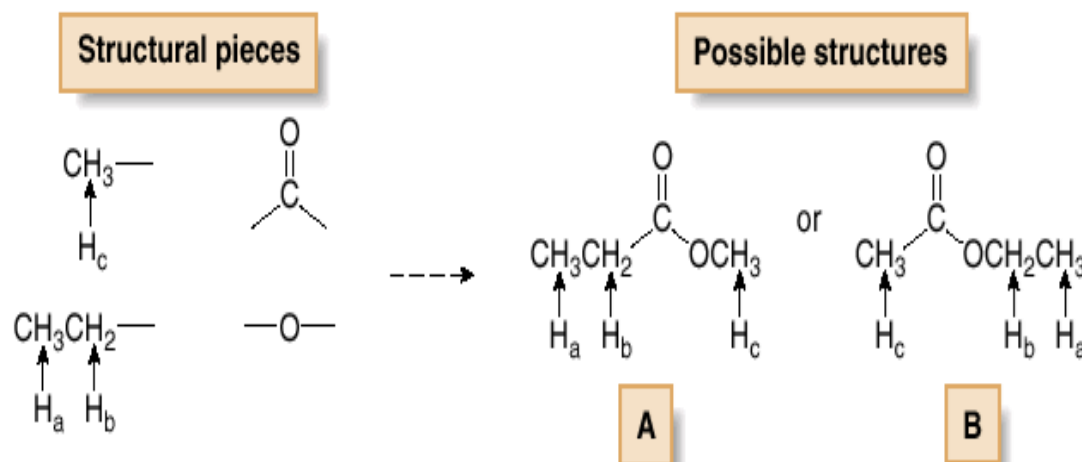


To summarize, **X** contains CH_3- , CH_3CH_2- , and $\text{C}=\text{O}$ (from the IR). Comparing these atoms with the molecular formula shows that one O atom is missing. Because O atoms do not absorb in a ^1H NMR spectrum, their presence can only be inferred by examining the chemical shift of protons near them. O atoms are more electronegative than C, thus deshielding nearby protons, and shifting their absorption downfield.

How To, continued . . .

Step [4] Use chemical shift data to complete the structure.

- Put the structure together in a manner that preserves the splitting data and is consistent with the reported chemical shifts.
- In this example, two isomeric structures (**A** and **B**) are possible for **X** considering the splitting data only:



- Chemical shift information distinguishes the two possibilities. The electronegative O atom deshields adjacent H's, shifting them downfield between 3 and 4 ppm. If **A** is the correct structure, the singlet due to the CH₃ group (H_c) should occur downfield, whereas if **B** is the correct structure, the quartet due to the CH₂ group (H_b) should occur downfield.
- Because the NMR of **X** has a singlet (not a quartet) at 3.7, **A is the correct structure**.

Applications of NMR Spectrum

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- Identification of structural isomers
- Detection of hydrogen bonding
- Detection of aromaticity
- Distinction between Cis-trans isomers and conformers
- Detection of electronegative atoms or group
- Detection of some double bond character due to resonance

References:-

- Elementary Organic Chemistry, Principles and Chemical Applications; by Y.R. Sharma, Pg.no: 180 – 237.
- Basics Concepts in Pharmaceutical Research, by Bijiya Ghosh and Sonal Dubey; Pg. no: 160 – 175.

Thank You